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Statistical and AI-Based Techniques for Badminton Data Processing: Concepts, Applications, and Challenges

Técnicas estadísticas y basadas en IA para el procesamiento de datos de bádminton: conceptos, aplicaciones y desafíos

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Abstract

Recent advances of artificial intelligence, data science, sensors and wearable technologies have fostered the emergence of an increasing number of tools, services and approaches for collecting and processing sports data by promising information that can be used to improve athletes' performance in a variety of contexts. Recent sports research publications indicate a trend on topics related to artificial intelligence (AI) and computer vision for data acquisition and analysis. However, this trend also raises questions from practitioners and researchers that are not familiar with the underlying concepts and technologies of such tools and approaches since their experience and education are mostly focused on the sports domain. Some of such questions are: what is the difference between these aforementioned technologies and techniques? What are common mistakes and caveats when trying to apply them in practice? Which of them better fits to overcome the challenges related to sports science from data collection to its analysis? With those questions in mind, this paper aims to provide an overview of the fundamental concepts of artificial intelligence and statistical data analysis, with a particular focus on their application to badminton-related challenges. We cover from classic statistics-based analysis to state-of-the-art deep learning models, addressing issues such as data acquisition and processing, and the limitations of each technique. We also discuss the interpretation of complex data outputs, since users must be aware of the limitations and potential biases of the algorithms to ensure that the insights provided by the results are relevant. We aim this knowledge can empower sports professionals to make informed decisions and effectively leverage technology to improve athletic performance and sports organizations.

Keywords: *Badminton, artificial intelligence, statistical analysis, sports analytics, data analysis.*

Resumen

Los recientes avances de la inteligencia artificial, la ciencia de datos, los sensores y las tecnologías portátiles han fomentado la aparición de un número creciente de herramientas, servicios y enfoques para recopilar y procesar datos deportivos que prometen información para mejorar el rendimiento de los atletas en una variedad de contextos. Publicaciones recientes en investigación deportiva señalan una tendencia hacia temas relacionados con inteligencia

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artificial (IA) y visión artificial para la recolección y análisis de datos. Sin embargo, esta tendencia también plantea preguntas de profesionales e investigadores que no están familiarizados con los conceptos y tecnologías subyacentes de tales herramientas y enfoques, ya que su experiencia y educación se centran principalmente en el dominio deportivo. Algunas de ellas son: ¿cuál es la diferencia entre estas tecnologías y técnicas? ¿Cuáles son los errores más comunes y recomendaciones al tratar de usarlas en la práctica? ¿Cuál de ellas encaja mejor para superar los retos relacionados con la ciencia del deporte desde la recolección de datos hasta su análisis? Con estas preguntas en mente, este documento tiene como objetivo proporcionar una visión general de los conceptos fundamentales de la inteligencia artificial y el análisis de datos estadísticos, con un enfoque particular en su aplicación a los desafíos del bádminton. Abarcamos desde el análisis clásico basado en estadísticas hasta modelos de aprendizaje profundo de última generación, abordando cuestiones como la recolección y el procesamiento de datos y las limitaciones de cada técnica. También discutimos la interpretación de los resultados de datos complejos, ya que los usuarios deben ser conscientes de las limitaciones y los posibles sesgos de los algoritmos para garantizar que los detalles revelados por los resultados sean relevantes. Nuestro objetivo es que este conocimiento pueda capacitar a los profesionales del deporte para tomar decisiones informadas y aprovechar eficazmente la tecnología para mejorar el rendimiento deportivo y las organizaciones deportivas. .

Palabras clave: *bádminton, inteligencia artificial, análisis estadístico, análisis deportivo, análisis de datos.*

INTRODUCTION

Data analysis plays a crucial role in contemporary badminton, enabling coaches, athletes, and analysts to make informed decisions based on empirical data. With the advent of advanced data collection technologies, it has become possible to identify and compare performance patterns, as well as optimize game strategies. These methods range from traditional statistical approaches to modern deep learning models, each offering unique advantages for understanding sports performance, particularly high dynamic sports like badminton.

This paper begins by discussing traditional statistical methods that serve as foundational tools for summarizing data and interpreting behaviors in sports analysis. Both descriptive and inferential statistics are crucial in this context; descriptive statistics such as averages and distributions facilitate the evaluation of performance metrics, while inferential statistics, including t-tests and ANOVA, allow for drawing conclusions from sample data and making comparisons across groups (Hastie et al., 2009). For example, based on temporal and spatial measures extracted from video footage, researchers can apply statistical methods to compare playing styles and strategies across different groups, such as men's and women's games (Abian-Vicen et al., 2013; Fernandez-Fernandez et al., 2013; Iizuka et al., 2020), junior and master players (Green et al., 2023; Fernandez-Fernandez et al., 2013), and Olympic Games editions (Abián-Vicén et al., 2018; Torres-Luque et al., 2019).

The increasing complexity of data — attributed to advancements in high-definition imaging technologies, wearable sensors, and GNSS-based tracking systems (such as GPS) — has led to a significant increase in the use of machine learning and deep learning techniques in sports analysis. Supervised learning algorithms, such as Support Vector Machines (SVMs) and Random Forests, have been applied to player evaluation, stroke type classification and match outcome prediction (Sinadia and Murwantara 2022; Wang et al., 2018; Sharma et al., 2021), for example. However, while effective, these algorithms often depend on simplified data representations, which may limit their applicability. In contrast, deep learning models, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), automatically detect complex temporal and spatial patterns from video data (LeCun et al., 2015). These models have been successfully employed to detect and track shuttlecocks, classify player movements and shot types from badminton match recordings and evaluate players' technical and tactical performance, for example (Cao et al., 2021; Rahmad et al., 2020; Luo et al., 2022). Information produced by such models enables coaches and athletes to refine strategies and enhance training methods based on detailed performance insights.

Whether employing statistical methods or AI-driven techniques, several essential factors must be addressed to ensure the validity and reliability of data analysis approaches. First, the foundation of any effective analysis lies in the quality and representativeness of the collected data. Key considerations here include the size and representativeness of samples, as small or biased datasets can lead to misleading results. In this context, accurate data acquisition protocols are essential to acquire reliable data and enable meaningful comparisons across studies. Furthermore, the availability and quality of public datasets are also critical for comprehensive analyses and reproducibility of the results.

Equally important is the recognition and minimization of biases in the analysis, particularly when using complex models. Performance metrics can be significantly influenced by contextual factors like training conditions, competition levels, and athlete demographics. By addressing these influences, researchers can enhance the robustness of their analyses and improve understanding of athletic performance.

In this paper, we will conduct a state-of-the-art review with the following objectives: (a) describing current applications of computational data analysis tools and techniques in the context of badminton, covering from basic statistics to advanced deep learning methodologies, and (b) providing concrete examples that allow readers to relate the presented concepts to their day-to-day work with badminton and other sports. We aim to reduce the gap between Data scientists and Sports Science researchers so they can achieve meaningful results from the data analysis tools hereby discussed.

STATISTICS-BASED APPROACHES

Statistical analysis is an essential component in the interpretation and understanding of collected data, offering a way to quantify performance, compare different variables, and identify patterns and trends. In the context of badminton research, where data can range from simple rally metrics to complex movement and physiological analyses, as well as subjective data collected through open-ended questions of survey forms, using appropriate statistical tools is crucial for deriving meaningful insights.

The choice of statistical tools should be guided by the research question, the nature of the data, and the assumptions underlying each method. In this section, we present a discussion on the statistical tools commonly employed in the analysis of badminton data. The aim is to provide a structured overview that not only clarifies the utility of each tool but also explores their practical applications in real-world scenarios. Additionally, we will reference key studies in the field, highlighting how these statistical methods have been effectively applied in badminton research to indicate meaningful insights, guide decision-making, and improve training strategies, for example.

Descriptive statistics

Descriptive statistics are foundational tools in data analysis, allowing researchers to understand the general structure of a dataset and identify overall patterns within the data before further analysis. Measures of central tendency, such as mean, median, and mode, help researchers to identify the most common values in a dataset, while range, variance, and standard deviation are essential for understanding the spread of data points and provide insights into variability and distribution. Additionally, the coefficient of variation is used to compare the dispersion across datasets with different units or scales (Hastie et al., 2009; Bindu et al., 2019).

The mean and standard deviation are often used to compare key badminton performance metrics across different genders or tournaments, such as the average match duration and shots per rally (Abián-Vicén et al., 2018; Gomez et al., 2019). These measures can also provide insights into the pace and intensity of matches, revealing patterns of performance variation when comparing temporal and stroke variables across tournament stages, game numbers, and match types (Abian-Vicen et al., 2013; Gomez et al., 2020; Torres-Luque et al., 2019; Gómez et al., 2021; Chiminazzo et al., 2018). For example, various studies have calculated the mean and standard deviation to facilitate comparative analyses of Olympic badminton matches over time (Laffaye et al., 2015). More specifically, Abián-Vicén et al. (2018) showed that badminton in the men's and women's doubles events evolved from the 2008 Beijing Olympics to the 2016 Rio Olympics towards longer matches with greater rest intervals between points. These conclusions were drawn from the analysis of key statistical measures, including mean and standard deviation. For instance, the women's doubles event saw an increase in real time played, with the average duration rising from 767.7 (± 242.2) seconds in 2008 to 913.5 (± 240.5) seconds in 2016, reflecting longer matches on average. Similarly, in the men's doubles, rest times grew from 21.4 (± 2.2) seconds in 2008 to 30.0 (± 5.1) seconds in 2016, indicating an increase in recovery periods. This suggests that, over time, players may have adopted more diverse or individualized recovery strategies, possibly in response to evolving match intensity and competition demands.

Dispersion measures play a crucial role in revealing the underlying uncertainty and consistency within data. Cabello-Manrique and González-Badillo (2003) highlight how the significant variability in certain metrics, such as performance times and lactate concentrations, reflect the unpredictable nature of badminton. In contrast, other metrics, like maximum heart rate during the match and the ratio of shots, show greater consistency, reflecting the stable physiological demands of the sport. By computing the coefficient of variation, the authors were able to normalize this dispersion, providing a clearer comparison of relative variability across different datasets, even when these datasets are on different scales. Chiminazzo et al. (2018) found differences between the group and play-off phases in 2016 Olympic Games regarding total time, total rally duration and total rest duration.

Other metrics, not based on the mean and standard deviation, are less commonly used, but can provide a better assessment of data variability, especially in cases where traditional measures may not fully capture the underlying distribution, e.g. in the presence of outliers. Gómez-Ruano et al. (2020) and Gomez et al. (2020) used the median to analyze temporal parameters such as set duration, number of strokes per rally, and factors related to long rallies, providing insights into the centrality of time-related demands. Additionally, lower and upper quartiles were calculated by Gómez-Ruano et al. (2020) to highlight the relative standing of data points. The median was also used to analyze the point difference established by winners and losers in badminton matches (Barreira et al., 2016). Although the mode is less frequently used, it helps identify the most common actions in a game. Skewness and kurtosis, as applied by Ali and Nguang (2023), offer further insights into the data distribution, identifying outliers that can impact badminton performance analysis.

Descriptive statistics are essential for summarizing and understanding the fundamental characteristics of a dataset. Through them, researchers can extract valuable insights into player performance, match dynamics, and factors influencing outcomes in badminton, facilitating the evaluation of how different competitive contexts affect player performance and guiding the design of specific training sequences. However, it is important to note that descriptive statistics are limited in scope; they can describe data but cannot provide insights into the significance or underlying relationships between variables. They also offer no means to test hypotheses or make predictions beyond the observed data. This is where inferential statistics, the focus of the next section, can be applied (Janczyk and Pfister 2023).

Inferential statistics

Inferential statistics extend the organization of data provided by descriptive statistics, allowing researchers to draw conclusions about a population from a sample by determining whether observed patterns are genuine or merely the result of random variation. In badminton research, inferential statistics can be used, for example, to evaluate whether performance differences between male and female players, or between early and late tournament stages, are consistent across tournaments or or simply reflect random variation (Abián-Vicén et al., 2018; Chiminazzo et al., 2018; Fernandez-Fernandez et al., 2013).

Significance, a central concept indicating the likelihood that a result is not due to chance, can be assessed using methods like t-tests and Analysis of Variance (ANOVA), which are widely applied to determine statistically significant differences between groups. T-tests provide a straightforward approach to hypothesis testing, being used to evaluate whether there is enough evidence in sample data to support a specific claim about a population — for example, comparing the mean performance of two badminton players under different training regimens. ANOVA extends that to compare means across three or more groups, e.g. when analyzing the effectiveness of multiple training techniques on players' agility. To analyze relationships between categorical variables, such as preferred playing strategies, Chi-square tests are employed to assess associations between categories (Janczyk and Pfister 2023).

Gomez et al. (2019) used a Student's t-test to analyze gender differences for variables and each match context. Additionally, they also computed Cohen's effect size — a technique that allows quantifying the size of the difference between two means, offering insights into the practical significance of the observed differences. To better understand the relationships between categorical variables, such as gender, range of strokes, and point outcomes, the authors utilized Chi-square tests. Furthermore, they computed Cramer's V, a measure that quantifies the strength of observed associations. Fernandez-Fernandez et al. (2013) computed pairwise t-tests to independently assess the effect of gender and set on variables such as rally duration, rest time and strokes per rally. The results showed differences between genders, with male players having longer rallies, executing more strokes per rally, and resting more between rallies than female players. These findings are relevant when designing training programs, for example.

ANOVA was used to compare physical performance metrics of Badminton players across multiple tournaments (Abián-Vicén et al., 2018; Laffaye et al., 2015) or tournament stages (Iizuka et al., 2020), as well as differences in motion kinematics (Cui et al., 2022) or shuttlecock speed (King et al., 2020) between different groups. The specific type of ANOVA used — whether one-way, three-way, or Multivariate ANOVA (MANOVA) – depends on the number of independent variables and the complexity of the analysis. Gomez et al. (2020) used a one-way ANOVA to analyze the time between strokes among competition clusters, such as long rallies. Gómez et al. (2021) employed a three-way ANOVA to explore the effects of tournament stage, game number, and game half on multiple dependent variables, including match scoring coordination, standard entropy, rally time, rest time, density, and racket-to-racket time, providing insights into the complex dynamics of badminton matches. Abian-Vicen et al. (2013) utilized a two-way ANOVA with game and event as factors, and for variables like match duration, rest time, and percentage of time played, they applied a Student's t-test for independent samples to analyze gender differences. Abián-Vicén et al. (2018) used a two-way ANOVA to evaluate differences in the temporal and notational dynamics of men's and women's doubles badminton matches across the 2008 Beijing, 2012 London, and 2016 Rio Olympic Games.

Before applying ANOVA or t-tests, data should meet the assumptions of normality and homogeneity (Hastie et al., 2009) to ensure the validity of the analysis results. If these assumptions are not met, the validity of the results may be compromised, potentially leading to misinterpretations. Normality tests, such as the Kolmogorov-Smirnov (Fernandez-Fernandez et al., 2013; Marchena-Rodriguez et al., 2020) or Shapiro-Wilk (Iizuka et al., 2020; Chiminazzo et al., 2018) tests, are commonly used for this purpose.

It is important to recognize the limitations of normality tests, particularly in studies with small sample sizes, which are common in sports research (Verma and Verma 2020). Small sample sizes can reduce the power of these tests, making it difficult to detect deviations from normality. When data fail to meet the normality assumption, non-parametric methods, such as the Mann-Whitney U test (Green et al., 2023; Chiminazzo et al., 2018) or the Kruskal-Wallis test, are recommended as robust alternatives. These tests do not rely on the assumption of normal distribution and can offer more accurate results under such conditions (Janczyk and Pfister 2023).

It is important to note that these approaches can be combined to strengthen the robustness of the findings and provide more detailed insights into the factors under study, as demonstrated by Green et al. (2023) and Gomez et al. (2020), for example.

Limitations and considerations

While statistical approaches provide valuable insights, their limitations must be carefully addressed to ensure that conclusions are both statistically valid and meaningful in the context of badminton research. Since statistical measures are typically derived from samples rather than entire populations, the reliability of the derived estimates depends heavily on the sample's size and representativeness (Verma and Verma 2020). For instance, suppose we need to calculate the mean height of Brazilian badminton players. Relying on a biased sample drawn from a single club or region may fail to capture the diversity of heights influenced by regional, socioeconomic, or demographic factors. Also, a small sample, such as 20 players, is unlikely to reflect the true variability present in the broader population, leading to potentially misleading conclusions. Similarly, inconsistencies in data acquisition, such as variations in the collected variables or differences in measurement protocols, can further compromise the reliability of analyses.

When interpreting statistical results, it is crucial to consider the context of the data, including factors such as the competitive level, the athletes' playing style, and the specific conditions of data collection. Care should be taken not to overgeneralize findings that may be strongly linked to particular circumstances, such as training environments versus real competitions. In other words, both the significance of results and the adequacy of sample size must be carefully considered to ensure that conclusions are reliable and applicable beyond the specific dataset. For example, stroke patterns observed in elite Olympic athletes may not directly apply to the training of youth players due to differences in skill level, physical attributes, and experience. Thus, insights derived from statistical analysis must be contextualized to ensure relevance and avoid misleading conclusions.

Moreover, when dealing with sports data, which frequently exhibit temporal and spatial correlations, traditional statistical methods often struggle to address more complex aspects, such as tactical patterns and dynamic game strategies. For more robust insights and in order to avoid biased interpretations, it is necessary to integrate statistical approaches with complementary techniques, such as machine learning or multivariate models, which allow for exploring nonlinear relationships and capturing finer performance variations. Data science techniques also offer a powerful approach, allowing researchers to uncover patterns, trends, and outliers that reveal interactions and patterns that might otherwise be overlooked in conventional statistical assessments. In fact, it is important to emphasize that all machine learning models — responsible for the recent surge of "artificial intelligence" solutions — are statistical in nature (it can be argued that they are in fact very large/complex statistical models (Goodfellow et al., 2016)), meaning that they are often subject to the same risks as classic approaches: biases, overgeneralization based on small samples, mistaking correlation for causation, and drawing conclusions from an incomplete view of a problem. Therefore, a basic understanding of such concepts is essential before using AI-based solutions for analysis, as discussed in the next section.

ARTIFICIAL INTELLIGENCE

The term "artificial intelligence" (AI) has been present in several contexts — from academic circles to the gaming industry — for several decades (Minsky 1961; McCarthy 1981; Funge 1999), but recent breakthroughs have considerably increased the public view of AI as a current technology, rather than a future promise seen in fiction. This led people to have different ideas about what "AI" really means, potentially resulting in miscommunication and unreal expectations. In this section, we aim to clarify the terminology in the AI field, allowing a clearer comprehension of the strengths and limitations of AI approaches when applied in practical racket sports scenarios, particularly in the badminton context.

First and foremost, we address the term "artificial intelligence" itself. For years, works of fiction have associated the term with robots and systems that mimic human intelligence, capable of reasoning, talking, seeing, and perceiving the world through various sensors and inputs such as cameras, microphones, or text. Research on the field may have departed from the same inspiration, but after initial attempts in the 1950s (McCarthy et al., 2006), particularly in the field of computer science, AI quickly started being used as an umbrella (and somewhat ill-defined) term for many different types of techniques.

The techniques responsible for the recent surge in AI applications for the general public are mostly covered by the subset labeled as "Machine Learning", or more specifically, its subset called "Deep Learning". Machine learning algorithms find patterns in the input data, producing "models" that can be then used to infer the properties of a new input, or to make predictions about sequences (Russell and Norvig 2020). In its simplest form, then, machine learning can be seen as an application of statistics concepts, but usually working on complex data with more dimensions, and models with more parameters, with computers being used to deal with the complexity and/or large volume of data ("big data"). As for the deep learning subset, it refers to recent machine learning approaches that are based on evolutions of a type of model called "artificial neural network", built from many simple processing units, stacked in layers, in a configuration called an "architecture". The "deep" aspect comes from the large number of layers that are usually stacked in modern architectures, whose processing has become possible in recent years due to advances in the use of GPUs (Graphical Processing Units) for parallel processing of great masses of data (Goodfellow et al., 2016). A more detailed study on deep learning architectures is beyond the scope of this paper.

AI techniques share the characteristic of being able to produce approximate solutions for problems whose optimal or exact solutions, via conventional algorithms, are unknown or widely considered unfeasible. For instance, consider the classic example of a chess-playing AI. A naïve optimal algorithm would take the current state of the board and predict every possible future move and board configuration to select the move that maximizes the probability (likelihood) of winning. However, such a solution is unfeasible, given the extremely high number of possible board configurations. Thus, a chess-playing AI would work on a probabilistic manner, based on heuristics (Russell and Norvig 2020) (usually in the form of sets of rules based on numeric scores for boards with certain properties), previously stored board configurations, examples from chess manuals that were programmed by a human, etc. The AI is not truly "intelligent" in the sense that it does not reason about chess or understand its rules, nor does it have awareness of the opponent or the context in which the game is being played — only the moves it chooses can be seen as "intelligent" in some sense.

The application of AI in racket sports, particularly badminton, presents distinct challenges and opportunities compared to traditional board games like chess. While chess is a discrete game with easily programmable rules and a finite, albeit large, number of possible states, badminton is fast-paced and dynamic, requiring AI to process continuous data streams and adapt to constantly evolving conditions on the court. To better understand this, consider a practical, albeit hypothetical, example: a system that receives as input a video of a badminton match and outputs a list of shot types made by the players in each rally, organized by set. The raw input consists of a sequence of images, each encoded as matrices that capture pixel intensities for every video frame. The number of possible pixel configurations across time is extremely large — equal to the number of possible videos that could exist. We have different tasks, such as: identifying the beginning and the end of each set, locating the player hitting the shuttlecock (as well as the shuttlecock itself), and determining the hit moment and type of the shot. How would a practical system deal with that?

In most approaches, the key lies in the so-called training phase, where machine learning algorithms are "taught'' to recognize important information in the pixels, whether in individual frames (e.g. identifying players) or sequences of frames (e.g. classifying shot types or tracking the shuttlecock). Depending on the machine learning approach used, this phase may require manually labeling data (these approaches are categorized as "supervised learning") — for example, human annotators might mark the coordinates of players, the shuttlecock, and the net, or tag specific shot events. Note that several statistical approaches also rely on this process, where manual data annotation plays a crucial role in model development. If obtaining labeled data is impractical or too time-consuming, patterns might be identified in the data without the need for manual annotations — for example, by employing clustering algorithms to automatically group similar shots or movements, in a type of approach called "unsupervised learning. After this phase, the trained model is expected to analyze unseen video footage. However, it is important to emphasize that success relies significantly on the representativeness of the training data — for example, performance under unseen conditions can be harmed if the training data has a limited number of environments, players, techniques, camera placements, etc.

The learning process can also be based on interactions with the environment: within a reinforcement learning (RL) framework, the system continuously refines its strategies by analyzing real-time feedback from match outcomes. Thus, instead of relying on labeled data or finding patterns in unlabeled data, RL focuses on trial and error. This process enables it to identify which actions lead to success or failure, allowing for adaptive improvements in decision-making over time and enhancing effectiveness in dynamic scenarios (Sutton and Barto 2018).

Another common practice is the use of existing generic datasets and pre-trained models. In our previous example, a pre-trained model for human detection could be integrated into a badminton AI system to efficiently identify players, while customized models should be developed for recognizing shot types, predicting player movements, or identifying tactical patterns. Note that a system specialized for a badminton context would still require careful consideration about several other aspects, since it is not possible to rely on generic pre-trained models for every single task, and the rules of the game and relevant elements are not programmed or informed to the system, they are learned from examples (Weiss et al., 2016).

In the following sections, we will explore different learning paradigms, outlining key considerations when applying these approaches. It's important to stress that these methods are not one-size-fits-all solutions — success requires a robust and representative dataset, and models cannot be expected to perform well without careful design and adaptation, for example. In addition to the main concepts, we also discuss works that use these methods to solve problems in the context of badminton.

Traditional Machine Learning

This section discusses the main principles and practical applications of classical machine learning approaches, specifically focusing on the supervised, unsupervised, and reinforcement learning paradigms, highlighting how they complement each other in addressing diverse challenges in sports data analysis. All the methods we discuss here involve two stages: training and inference. During the training stage, the model receives examples of inputs and produces a computational model that represents and generalizes the training data. During the inference stage, the trained model makes predictions and produces the outputs for previously unseen inputs. When a system is under development, the inference stage is also called the test stage, with the model receiving a set of samples that were not used during training, and the produced outputs being compared to the expected results to measure the model's prediction performance.

Supervised Learning

Supervised learning is the most widely used type of machine learning approach (Russell and Norvig 2020). During the training stage, the model is provided with a set of labeled data samples, which consist of examples of inputs the system will receive, along with their expected outputs. For example, for a system that detects and tracks the shuttlecock, the input could contain videos of badminton matches, with the labels being the coordinates of the shuttlecock in each video frame. In another example, for a system that relates player physical attributes (weight, height, age, gender, etc.) with performance, the inputs would be lists of attributes from each player, and the labels of the associated numeric performance indicators. Figure 1 illustrates the workflow of supervised learning, where the process involves training a model to detect shuttlecocks and players in images.

Supervised learning approaches differ between each other in several aspects: how the raw data is processed for internal representation (usually through a process called "feature extraction"); how the universe of possible processed inputs (called the "feature space") is analyzed, explored and divided; and how the model is updated during the learning process; among other characteristics. Next, we will explore some supervised learning approaches found in the literature, particularly of badminton studies.

Ghosh et al. (2018) proposed a framework for automatic attribute tagging and analysis of badminton videos. First, to differentiate between "point" and "non-point" frames, they extracted features from video frames using the Histogram of Oriented Gradients (HOG) algorithm, which captures details related to edges and texture. To classify these frames, they employed a Support Vector Machine (SVM), a robust classification algorithm well-suited for handling complex, high-dimensional data. HOG features were also employed to segment player strokes, serving as inputs for a deep learning-based approach. Additionally, the study also addressed player identification and tracking.

Supervised learning techniques have been applied to player evaluation in badminton using various types of data. For instance, Dieu et al. (2020) used an SVM-based algorithm to classify player expertise, based on rally duration, time between rallies (collected manually), and accelerometer data from simulated matches. Sinadia and Murwantara (2022) compared machine learning models for athlete profiling using official match scores from 2018 to 2021, finding that Random Forests outperformed SVMs when predicting the value of a continuous variable, but yield similar performance when assigning discrete classes. Random Forests are ensembles built of multiple decision trees, each one being responsible to partition data

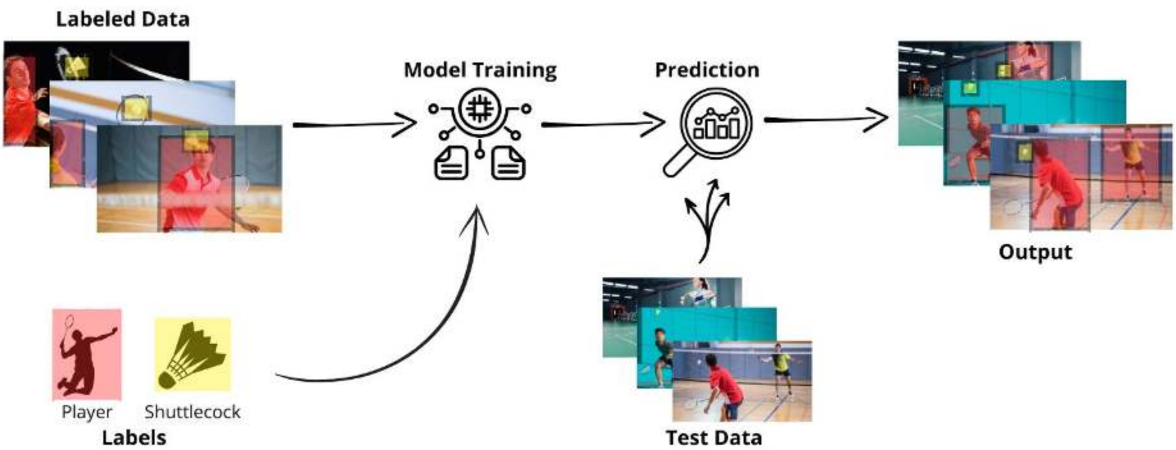

Figure 1. Supervised learning approach

into hierarchical subsets, making them particularly effective for handling categorical data. Random Forests merge the predictions from multiple decision trees, and are typically robust to noise due to its ability to capture complex, non-linear relationships in data (Bishop 2006). Additionally, Ghosh et al. (2020) proposed StanceScorer, a system that evaluates player performance based on stance and posture using wearable device data. To compare players' stances with those of professionals, they explored machine learning techniques such as k-Nearest Neighbors (k-NN), a classification algorithm that bases its predictions on the proximity of a data point to its nearest neighbors in the feature space.

Wang et al. (2018) also explored data from wearable devices, but focused on the classification of three types of strokes: smashes, clears, and drops. To optimize performance and minimize bandwidth requirements during communication with a cloud server, they pre-processed the data using Principal Component Analysis (PCA), a technique that simplifies complex datasets by reducing their dimensions while retaining essential information. Stroke labeling was integrated with data acquisition, as athletes were instructed to perform the strokes in a predefined sequence. The authors compared the performance of different classifiers, including SVM and k-NN. Additionally, they also classified athletes by skill level (elite, subelite, and amateur) based on wrist motion patterns, showing that elite players exhibited more consistent motions. Alternatively, Ramasinghe et al. (2014) proposed an automatic stroke recognition method for low-resolution broadcast videos, based on dense trajectories and trajectory-aligned HOG features. An SVM classifier was employed to categorize strokes into four types: smash, forehand, backhand, or other.

Supervised learning methods were also employed for match outcome prediction in badminton, as in Sharma et al. (2021). The study used raw statistical data from various tournaments held between 2016 and 2019, collected from the official Badminton World Federation (BWF) website and stored in CSV format. After the data cleaning and transformation steps, relevant features were selected to train several machine learning models. The authors achieved up to 100% accuracy in predicting match outcomes for specific tournaments. For example, using Naïve Bayes with Correlation-Based Feature Weighting, the model accurately predicted the results of all matches in the men's singles, men's doubles, women's singles, and women's doubles categories during the German Open 2019 Championship.

Other supervised learning applications in the context of badminton include shuttlecock tracking and trajectory estimation from video (Lee, 2016), and tactical pattern analysis (Coenga, 2023). For instance, to explore the association between a player's position during the conclusion of a point and the related characteristics, such as shot type and impact area, Coenga (2023) proposed a decision tree-based model.

Unsupervised Learning

Although supervised learning techniques have proven effective in various applications, they often require extensive labeled datasets, which can be challenging to obtain. Labeling data accurately can be time-consuming and may introduce biases, especially in complex sports environments where player actions vary widely. Moreover, supervised learning can fail to capture complex relationships or non-linear patterns in the data. These limitations highlight the importance of unsupervised learning methods, which do not rely on labeled data, focusing on finding patterns or structures within the data itself. During training, the system processes unlabeled data to identify inherent relationships, such as clusters or correlations, without predefined targets. The inference stage, as in supervised learning, involves applying these learned patterns to new data to reveal insights about the data's underlying structure (Bishop 2006). Note that, although unsupervised learning can succeed in identifying patterns without relying on manually labeled samples, this task is considered challenging, and supervised learning remains as the most common choice when possible.

Gómez-Ruano et al. (2020) analyzed the serving performances of badminton Olympic medalists using statistical approaches, along with cluster analysis and decision tree methods. The study revealed that medalists demonstrated a greater variety of serve types compared to their opponents, which significantly influenced their point outcomes and overall match dynamics. In their study, Ding et al. (2024) determined the optimal positioning for a player using unsupervised clustering techniques, specifically hierarchical clustering. They grouped various locations into clusters to derive a recommended position for the receiving player, which enhances the probability of successfully controlling the shuttlecock.

K-means clustering has proven to be an effective tool for analyzing athlete performance and identifying distinct patterns within a group. For instance, Sinadia and Murwantara (2022) applied clustering models to categorize badminton players' game performances into groups that distinguish different levels of athletic ability based on game results. By employing the k-means method, they successfully identified separate clusters representing players with similar motor capacities. Similarly, Pan et al. (2024) applied various methods, including k-means clustering, to explore the neuromuscular control strategies of advanced and beginner badminton players executing jump smashes.

Reinforcement Learning

In addition to supervised and unsupervised learning, it is also possible to consider reinforcement learning, a unique learning paradigm in which an agent learns optimal decision-making strategies through interaction with an environment. Instead of relying on labeled data or pattern discovery in unlabeled data, reinforcement learning focuses on trial and error. At each time step, the agent takes an action, receiving feedback in the form of rewards or penalties, which modify the state of the environment. The agent's objective is to learn a near-optimal policy that maximizes cumulative long-term rewards by refining its actions based on the experienced consequences. The relationship between the agent and the environment, along with the key elements of action, state, and reward, collectively defines the reinforcement learning framework, making it particularly effective in complex decision-making scenarios such as game playing (Sutton and Barto 2018).

Huang et al. (2023) presents a reinforcement learning environment for badminton based on the Multi-Agent Particle Environment, developed to simulate realistic tactical play between two agents in a rally. With multi-view perspectives (side and top views) for detailed observation, the environment allows agents to decide on shot type, target position, and defensive positioning in a turn-based format, focusing on tactical learning rather than real-time speed. To improve realism, meta-parameters such as player speed, defensive range, and shot trajectory physics are calibrated from match data, ensuring simulations compatible with actual match dynamics. Liu et al. (2013) proposed an approach based on reinforcement learning to time-optimal control of a badminton robot during the serve operation. The racket must move from a resting position to a target state with specific position and velocity constraints. In the experiments, the robot achieved desired motion patterns in approximately 200 trials. The study further compares a model-based optimization, analyzing control performance, convergence properties, and the respective advantages and limitations of each method.

Traditional supervised, unsupervised and reinforcement learning approaches are employed in a wide range of contexts, from player evaluation to shuttlecock tracking. These methods can use diverse data sources, feature types, and extraction techniques (for example, manual video annotations, publicly available match data, and even wearable device data). However, supervised learning approaches still tend to produce simpler and more robust models. Although such techniques continue to be applied and frequently cited in the literature, their role in badminton-related research is gradually being replaced by more advanced alternatives: deep learning models. This change is related to broader trends in data analysis, where increasingly complex tasks demand more flexible and scalable models.

Deep Learning

The majority of recent works and applications related to “artificial intelligence” are based on deep learning approaches. Deep learning is a subset of the machine learning field, and as such, the same principles apply: concepts such as supervised and unsupervised learning, the separation into training and inference/test stages, the data-driven approach to produce models, etc. Deep learning derives from previous works on artificial neural networks (Russell and Norvig 2020), and is characterized by complex architectures with multiple stacked layers (LeCun et al., 2015). An architecture is a particular organization of simpler processing blocks. State-of-the-art architectures can be very large and even contain entire older architectures within them.

The greater complexity of deep learning when compared to more traditional approaches brings both advantages and disadvantages. On the one hand, it allows modeling complex, high-dimensional data and relations, achieving results that are not feasible with older approaches. On the other hand, all that complexity demands huge amounts of processing power and data — in fact, training very large models in recent years has become unfeasible for small teams and individual researchers, becoming the domain of large companies. There is still ongoing research focusing on dealing with these disadvantages; seeking smaller, more specialized models; and exploring existing models for extracting information from specific problem domains — this last topic is of particular importance for our discussion, since most research applying deep learning in a badminton context fits this description. In the following, we explore some of these works.

Various studies have explored shuttlecock detection and tracking, a challenging task due to its high-speed movement. Huang et al. (2019) introduced TrackNet, a deep learning network designed to track shuttlecock trajectories under challenging conditions. The model employs a popular network called VGG16 for feature extraction, followed by an upsampling network, to generate a heatmap that represents the probability distribution of the shuttlecock's center, using data from several consecutive video frames. TrackNetV2 is an enhanced version proposed by Sun et al. (2020), who introduced improvements in processing speed, accuracy, and memory efficiency — for example, TrackNetV2 incorporates a U-Net (a popular image segmentation network) into its architecture and optimizes GPU memory usage. Similarly, Cao et al. (2021) proposed two variants of Tiny YOLOv2, a well-known deep learning-based detector, specifically adapted for shuttlecock detection. The system also tracks trajectories, providing data that allows a robot to position itself and use a mechanical arm to return the shuttlecock to the opponent. He and Zhang (2024) predicted trajectories using another version of the Tiny YOLOv2 network, incorporating structural and loss function modifications, combined with an Unscented Kalman Filter for tracking. Liu and Wang (2022) focused on the extraction and segmentation of 3D shuttlecock trajectories from monocular videos by combining badminton court dimensions and physics with advanced tracking techniques. The authors used an architecture inspired by TrackNet to predict shuttle masks, which are images that clearly show where the shuttlecock is located in each frame, across three consecutive frames. Additionally, a recurrent network model integrates court layouts, player poses, and shuttlecock positions to predict hits.

Chou et al. (2024) explored event detection in badminton match videos, focusing on precise shuttlecock detection to support tasks such as hit counting and timing. A two-step process was adopted: the initial object detection model focuses on the shuttlecock and player locations, while a subsequent post-processing model refines the predictions for each task. Since the initial detection accuracy using TrackNet was insufficient, the authors implemented various deep learning methods to enhance precision when dealing with noisy data. They used YOLOv7 for initial object detection and an Asymmetric U-Net for improved shuttlecock detection. Post-processing uses a model called EfficientNet for refining predictions for specific tasks, such as player positioning and ball height.

Chien and Yu (2023) proposed an approach to detect hit-frames from video footage that integrates rally-wise video trimming, player and court keypoint detection, shuttlecock flying direction prediction, and hit-frame identification. By using CNNs and transformers (a type of deep learning architecture known for handling sequential data efficiently), the authors achieved expressive results in shot angle recognition and shuttlecock direction prediction based on player keypoints. Their tool identifies frames where players hit the shuttlecock, providing a foundation for detailed analyses of strokes and player movements, with potential applications in training optimization and strategy development.

Rahmad et al. (2020) investigated the use of pre-trained CNN models to automatically detect and classify badminton actions that represent types of strokes, including smash, clear, drop, lift, and net shots, from broadcasted videos. These pre-trained CNNs are models previously trained on large, general-purpose datasets, which are able to extract fundamental visual patterns, and which can be fine-tuned to detect specific actions in badminton videos. The authors compared the performance of four models — AlexNet, GoogleNet, VGG-16, and VGG-19 — and concluded that pre-trained models, especially AlexNet and GoogleNet, hold promise for automating notational analysis.

Zhao (2023) also proposed a method for temporal positioning and classification of shot motions in badminton videos, specifically focusing on the recognition of arm movements during shuttlecock shots. Their approach measures the player's arm movement to precisely locate shot timing and extract shot-sequences — video segments that contain the complete motion of a shot. Shot-sequences were classified as: forehand, backhand, smash, and drop shots. For finer classification within smash shots, image processing techniques are used to analyze the shuttlecock's motion trajectory, distinguishing between high clears and kill shots. In contrast, Steels et al. (2020) proposed a cost-effective and non-intrusive method for classifying badminton movements based on accelerometer and gyroscope data. This study defined a novel CNN architecture that classifies nine distinct activities: seven specific badminton strokes, displacements, and moments of rest. The paper explored various sensor placements and sampling frequencies, showing that even lower frequencies can maintain adequate accuracy for classifying fast movements. To improve activity recognition reliability, the authors explored a weighted ensemble learning approach to combine outputs from models with different frame sizes. Van Herbruggen et al. (2024) proposed a model based on Long Short-Term Memory (LSTM) networks, which utilizes Ultra-Wideband data in conjunction with classification outputs from shot recognition systems to identify up to 11 commonly used badminton strategies. LSTM is a specialized type of recurrent neural network (RNN) designed to process and predict sequential data by retaining important information over long time intervals.

Luo et al. (2022) presented a novel methodology for analyzing badminton players' footwork using deep learning for shoe localization, more specifically a VGG16 model. The proposed system utilizes binocular vision, capturing video data from two strategically positioned cameras to extract the 2D image coordinates of players' shoes. These coordinates are then transformed into 3D data, enabling a quantitative analysis of footwork trajectories. Similarly, Jannet et al. (2024) compared YOLOv8 and YOLOv9 for detecting and analyzing badminton players' footwork patterns. These approaches facilitate the analysis of inter-individual footwork adaptations during competitive play and their correlation with player performance.

In addition to classifying movements, it is equally important to be able to predict them. Wang et al. (2022) introduced ShuttleNet, a framework for stroke forecasting composed of two key components, both based on transformer architectures. One component models the progress of the rally, while the other isolates player-specific information to create individualized player contexts. These insights are then fused using a specialized neural network mechanism to integrate both the rally dynamics and player behaviors, allowing the model to predict the next shot type and the landing area of the shuttle. Alternatively, Chang et al. (2023) investigated both the prediction of returning stroke types and player movements in badminton matches. The authors introduced the Player Movements graph, which captures the strategic relationships between players and their locations during rallies. Based on this graph, they propose a novel model called Dynamic Graphs and Hierarchical Fusion for Movement Forecasting, which captures mutual interactions between players and their dynamic tactics over time, and incorporates hierarchical fusion modules that consider the influence of both players' styles. To classify rally outcomes, Tan et al. (2022) utilizes stroke representations derived from both HOG features and deep features extracted by a CNN, which are subsequently input into LSTM models.

Equally important is the need for higher-level analyses of technical and tactical performance in badminton, as understanding players' actions provides valuable insights into their strategies and effectiveness. In this context, Ding et al. (2022) used deep reinforcement learning techniques to evaluate players' technical and tactical performance. The proposed methodology analyzes not only game results but also the quality of player actions, such as stroke type and its correlation with scoring likelihood. The researchers employed LSTM networks, which are effective for capturing temporal dependencies in sequences — crucial for sports where action sequences impact rally outcomes. Using historical video data, they tracked player movements and shuttlecock positions using techniques like AlphaPose (Fang et al., 2017) and TrackNet (Huang et al., 2019) for precise pose estimation and tracking. Similarly, Wu and Chen (2023) proposed a technical analysis method based on an imaging system and calibration techniques for a binocular camera. This approach incorporates a dynamic identification algorithm using a sliding window, enabling real-time recognition of badminton hitting actions.

In another work focused on player action understanding, Ding et al. (2024) present a framework for analyzing teamwork in badminton doubles using a two-stream deep learning architecture. One stream processes player locations and velocities from the top view, while the other analyzes player poses from the back view. These inputs are combined and fed into an U-Net model to predict the control area probability map. The framework also employs a Gaussian Mixture Model for player positions and Graph Convolutional Networks for pose analysis. The study correlates scoring with control area and proposes practical applications for optimizing doubles positioning strategies. The paper also introduces the first annotated drone dataset for badminton doubles, including player bounding boxes, shuttlecock locations, and poses from top and back views.

Chen et al. (2023) developed an approach to analyze player behaviors for creating training and playing strategies based on opponents' weaknesses. After segmenting match videos into individual rallies, the authors use a ResNet-18-based neural network to classify frames as court or non-court views. Player positions are tracked using YOLO-v3, and poses are analyzed with OpenPose (Cao et al., 2021), with data transformed into a court coordinate system for precise analysis. The system also detects hit frames by training a classifier using player images and joint heat maps, combining W-ResNet-28 and LSTM networks for feature extraction and temporal correlation analysis.

Wang (2023) presents a novel closed-loop AI system designed to enhance badminton players' performance by addressing key challenges in leveraging deep learning within the sport. It outlines a comprehensive approach comprising six components: Badminton Data Acquisition, Imitating Players' Styles, Simulating Matches, Optimizing Strategies, Training Execution, and Real-World Competitions. Central to this framework is the development of RallyNet, a novel model designed to accurately replicate players' decision-making processes (Wang et al., 2024). Additionally, the authors also developed a sophisticated badminton simulation environment that incorporates real-world physics, enabling agents to emulate authentic game scenarios, including accurate shuttlecock trajectories and player interactions. This environment is based on the Multi-Agent Particle Environment and is enhanced with previous stroke forecasting models (Wang et al., 2022; Chang et al., 2023), providing both quantitative and qualitative results. By leveraging reinforcement learning techniques such as Deep Q-Network and Proximal Policy Optimization, the study optimizes player strategies while preserving their unique playing styles. The system continuously refines its strategies through trial and error, allowing for real-time adjustments based on feedback from simulated matches.

Limitations and considerations

The works cited in Sections *Supervised Learning* and *Unsupervised Learning* illustrate a trend towards increasingly complex models and solutions, with research distancing itself from classic procedures that have been relied upon for decades. For example, the idea of a human researcher watching hours of matches and videos while taking notes that will later fill a spreadsheet is giving place to fully automated solutions, able to collect raw information from video streams and even analyze actions and situations at a higher comprehension level.

Although this trend is certainly promising and observed in many other research fields beyond sports, it is important to stress the limitations of current deep learning and artificial intelligence tools. While more complex models can indeed deal with more complex data and situations, they also have high computational and energy demands, as well as the need for huge amounts of training data. This last aspect is perhaps the most critical, since if training datasets are not representative of the true variety of situations observed in practice, resulting models may be biased and fail to generalize properly when employed in practice. For example, if videos are recorded by a particular camera at a particular court, a model may fail when fed images from a different camera or place. The same can be said about actions from limited groups of people, and patterns observed on a certain group (e.g. olympic athletes) may differ from those of other groups (e.g. amateur players). Therefore, whenever a model is employed in a particular scenario, it must be first tested to check if its performance remains similar to what was initially reported.

One last point about the use of artificial intelligence tools: in recent years, applications based on large language models (LLMs) such as ChatGPT[1] became widely available, but their use in an actual sports setting needs some care. This type of tool can produce text about any subject, including badminton, its rules, history, strategies, etc., but only based on reorganizing content observed during training, and still with the possibility of "hallucination", a known issue of this type of model, where imprecise or incorrect information is presented due to confusion between similar structures and contexts (e.g. mentioning aspects from other sports when talking about badminton). Despite the impressive results that can be obtained from such tools, it is important to keep in mind that they do not have actual understanding about the sport, and cannot interpret a video so as to "think" about facts that could prove useful in practice. Therefore, as with every other tool presented in this section, LLMs could find some usefulness in extracting and presenting data, but human interpretation is still paramount, so ways of organizing, summarizing and presenting data to produce useful knowledge are very important.

Moreover, the rapid development of AI tools presents additional challenges related to ethical considerations, data privacy, and the accessibility of these technologies. The models must be carefully evaluated to ensure fairness, transparency, and reliability. Furthermore, interdisciplinary collaboration between computer scientists and sports experts is essential to properly align technological advances with practical needs and constraints, ensuring that the resulting models and systems are both accurate and applicable in real-world badminton settings.

In addition to these points, the use of AI in sport also involves broader ethical and practical constraints that influence how these tools can be adopted in different contexts. Systems trained on limited or unbalanced datasets may reproduce existing biases when applied to athletes with distinct characteristics or training environments, which reinforces the need for careful validation. Privacy remains a central issue as well, since video recordings, sensor traces and other personal data require clear consent procedures and appropriate safeguards. Finally, the deployment of AI depends on access to suitable infrastructure and technical expertise, which varies substantially across teams, institutions and countries. These disparities can limit the applicability of advanced methods in less-resourced settings and highlight the importance of solutions that are transparent, scalable and compatible with diverse operational realities.

In this context, effective collaboration between sports professionals and data specialists becomes central to ensuring that AI-based approaches are useful in practice. Coaches, analysts and other staff help define which performance questions are meaningful and how they relate to the realities of training and competition, while data scientists structure and analyze the information using appropriate methods. This continuous exchange also helps translate technical outputs into concepts that can be operationalized on court, reducing the gap between statistical indicators, model predictions and sport-specific terminology. As these tools are applied, expert feedback plays a key role in refining assumptions, adjusting metrics and identifying situations in which the model's behavior diverges from what practitioners observe. Maintaining this iterative, human-in-the-loop process supports the responsible integration of AI into badminton workflows, ensuring that automated analyses complement (rather than replace) professional judgement in real competitive settings.

DATA ACQUISITION AND TREATMENT IN BADMINTON

Effective data acquisition and processing are crucial for several tasks related to advanced analysis in badminton, including performance evaluation, strategy development, and athlete monitoring. Capturing a comprehensive range of scenarios that represent real-world conditions is essential, as that enables models to generalize effectively across diverse situations and player styles. Additionally, the quality, representativeness, and consistency of training data are crucial factors that directly affect the reliability of both machine learning models and statistical analyses. In the following, we explore key considerations in data acquisition, addressing essential aspects that researchers must take into account.

In badminton, various technologies are used for data collection, each offering unique insights into player performance and game dynamics. Motion sensors such as accelerometers and gyroscopes are frequently attached to rackets or worn by athletes to monitor stroke speed, attack angles, and body movements. These sensors provide detailed data on the dynamics of each movement, enabling precise analysis of players' technique and movement patterns, which can help identify areas for improvement, optimize performance, and reduce the risk of injury (Wang et al., 2018; Van Herbruggen et al., 2024; Seong et al., 2024). Physiological data such as heart rate, oxygen consumption, and lactate levels also provide valuable insights into endurance, recovery, and overall fitness when evaluating an athlete's physical condition and performance capacity, both during training and competition (Green et al., 2023; Fernandez-Fernandez et al., 2013). Video data offers a visual record of matches that can be used for sports data analysis, and is a powerful resource for understanding player behavior and game dynamics. In badminton, video footage typically comes from two primary sources: publicly available broadcasts, such as those provided by the Badminton World Federation (BWF), which offer high-quality footage with multiple camera angles, and privately captured recordings,

1 https://openai.com/chatgpt/

often made by coaches, analysts, and researchers for more targeted analysis (BWF TV 2024).

It is essential to follow standardized data acquisition protocols to ensure the reliability and consistency of data processing and analysis across various contexts. For example, having videos captured from a fixed camera position behind the court (as typically done by the BWF in official match recordings) facilitates the development of targeted algorithms — in contrast to the use of mobile cameras or those with unpredictable angles, which can introduce variability that require a more complex analysis and may even reduce the reliability of insights drawn from the video data. Similarly, sensor deployment must be standardized, clearly defining sensor locations or instructing participants to perform a consistent set of movements during data acquisition. In summary, standardized data acquisition protocols are essential, since without standards and documentation, data may become inconsistent, and differences in methodology and measurement criteria can complicate cross-study comparisons, limiting the ability to identify trends or draw conclusions that can be generalized. For example, if studies define or measure performance metrics differently, such as variations in the types of strokes or the specific attributes used for evaluation, the comparison of results may be distorted (Hastie et al., 2009).

Additionally, ensuring proper sampling is essential for obtaining valid and reliable results. This includes selecting a sample size that is both statistically significant and representative of the population. It is important to consider factors such as age, gender, and skill level, as these variables can significantly influence the outcomes and enable meaningful comparisons across different groups. For example, data collected from elite athletes, such as Olympic players, will likely differ from that of local, amateur players. Therefore, establishing clear benchmarks and ensuring that the sample reflects the target population is crucial for drawing valid conclusions (Verma and Verma 2020).

Once data is acquired, various processing stages may be required to prepare it for further analysis or visualization. This could involve formatting the data appropriately and aligning it with the specific objectives of the study, or even removing unnecessary data. For example, when using videos acquired by the BWF, the camera is typically positioned behind the court, but there are also close-up shots and different angles of key plays. Therefore, the initial step involves removing unnecessary frames that do not contribute to the analysis or do not follow a required standardization. Another pre-processing step could be adjusting the illumination conditions and applying noise reduction techniques, as varying lighting and noise might affect the accuracy of analysis (Gonzalez and Woods 1992).

In some cases, after extracting relevant features from the frames (such as shuttlecock positions, player movements, or trajectory patterns) it may be necessary to reduce the dimensionality of the resulting dataset. Techniques like Principal Component Analysis (PCA) can be employed to transform high-dimensional feature data into a more manageable form, retaining only the most significant components. This step is particularly useful when the extracted data contains a vast number of variables, many of which may be redundant or irrelevant to the study's goals. By simplifying the dataset, PCA helps to improve computational efficiency and focus the analysis on the most relevant features (Jolliffe and Cadima 2016).

Data labeling is also a concern. Supervised learning may require properly labeled data to ensure the model can learn from it, which can be done manually or using automated techniques. Manual labeling, often referred to as notational analysis, involves human reviewers assigning labels to specific events or patterns, such as identifying where the shuttlecock lands. This method, while accurate, is time-consuming and requires standardized protocols to maintain consistency across different datasets. Alternatively, automated approaches (using computer vision or machine learning algorithms) can significantly speed up the labeling process (Bishop 2006), although these approaches frequently need at least some amount of hand-labeled data before they can be used.

Furthermore, adopting standardized formats to store acquired data is critical to ensure data interoperability and promote collaboration across research initiatives. Using common formats, such as CSV and XML, reduces the time spent on data conversion and enhances compatibility between different datasets. This approach not only streamlines data sharing, but also increases the overall impact of research by making it easier to integrate data from diverse sources and create comprehensive large-scale studies.

Several studies have publicly shared badminton data. For example, Wang et al. (2023) introduced ShuttleSet, a dataset of stroke level records in single badminton, featuring 18 types of shots, player positions, and hitting locations, along with benchmarks for stroke influence, forecasting, and movement analysis. Ding et al. (2024) presented a men's doubles dataset filmed with dual 4K drones, ensuring complete coverage without occlusion. Similarly, Ban et al. (2022) offered a dataset for player-specific match analysis, including rally, stroke, and outcome annotations from nine top-level matches to support prediction and tactical studies. Van Herbruggen et al. (2024) developed a dataset with IMU (Inertial Measurement Unit) and UWB (Ultra-Wideband) data, capturing 13 distinct shots and 11 strategies performed by six professional and semiprofessional players. Huang et al. (2019) created a dataset from the 2018 Indonesia Open Final, containing 18,242 labeled frames with attributes like shuttlecock visibility and X,Y coordinates. Lastly, Sun et al. (2020) provided a dataset of 55,563 frames from 18 matches, including 15 professional and 3 amateur games, and Li et al. (2024) introduced the VideoBadminton dataset to improve action recognition in badminton. Open access to these datasets facilitates result verification, study replication, and further research, fostering transparency, improving robustness, and accelerating advancements in both academic and applied domains.

To complement these considerations, we include below a brief set of practical guidelines for researchers working with badminton data acquisition and preprocessing:

- Standardize data acquisition (video angles, lighting, and sensor placement) to ensure consistency across sessions; or adopt models and techniques capable of dealing with variations, with special attention to performance degradation due to changes in acquisition conditions.
- Record contextual metadata (player level, environment, equipment) that influences performance patterns.
- Apply preprocessing systematically, removing irrelevant frames and correcting illumination or noise issues.
- Use clear and consistent labeling protocols, especially for supervised learning tasks.
- Use standardized and widely compatible file formats to facilitate data sharing and integration with other studies.
- Document acquisition and preprocessing procedures to support transparency and reproducibility.

CONCLUSIONS

In this paper, we have discussed state-of-the-art approaches for badminton data analysis, also emphasizing key aspects of data acquisition and processing. Badminton data analysis involves different levels of granularity, including data from matches, players, and the shuttlecock. Match-level data, such as scores, rally durations, and point distributions, offer a view of performance trends. Player-specific metrics, ranging from basic statistics like the number of strokes per set to more intricate measures capturing court movement patterns during rallies, enable detailed performance evaluations. Additionally, shuttlecock data can provide insights into trajectory patterns, for example, which are critical for understanding game dynamics. Combining these levels of granularity is crucial, particularly in tactical analysis, where the interaction between players and the shuttlecock is critical.

Statistical methods play a central role in badminton data analysis, providing valuable insights by identifying trends, comparing performance metrics, and assessing variability in player actions. As for machine learning, it allows uncovering complex, non-linear relationships in the data, offering a deeper understanding beyond what traditional statistics can provide.

Data Science, although not yet widely explored in the context of badminton, can offer an exciting opportunity to bridge the gap between traditional statistical methods and more complex machine learning approaches. Unlike traditional statistics, which are often limited to descriptive summaries and hypothesis testing, data science integrates these methods within a framework that emphasizes discovering complex patterns and relationships. For instance, rather than just calculating averages or running t-tests, data science might use clustering to group players by performance trends or employ regression to model interactions between multiple variables across different game scenarios. This results in insights that are dynamic, exploratory, and applicable to real-world badminton contexts. While machine learning is one tool within data science, the latter also supports a flexible, accessible approach to data exploration, by incorporating simpler, interpretable methods that help uncover meaningful insights even in cases where predictions are not the primary goal. Data science can offer a practical solution for sports science professionals, enabling those without programming expertise to analyze data using tools like Python, R, and platforms such as Google Colab or Power BI. Unlike deep learning, which requires large datasets and specialized infrastructure, data science can be effectively applied to smaller datasets, like those often found in badminton. This allows professionals to uncover complex patterns, improve training methods, and gain insights into performance dynamics without the need for extensive resources typically required by more advanced techniques. Future research could explore how data science techniques can be applied to a variety of areas in badminton, from analyzing player performance and identifying patterns in match strategies to optimizing training regimens and improving tactical decision-making.

Limitations and Future Work

Building on potential directions for future work, we highlight several areas where AI could be explored more extensively. Injury prevention and rehabilitation could benefit from AI-driven models that analyze player movement patterns and predict potential risks, helping to reduce the likelihood of injuries. Automated match annotation is another promising area, where AI could streamline data collection, recording strokes, player movements, and rally durations with improved efficiency. Scouting and talent recruitment could also leverage predictive models to identify promising players in smaller tournaments by analyzing performance metrics and technical characteristics. Furthermore, training load evaluation, using wearable sensors and machine learning algorithms, could allow monitoring athletes' workload to optimize training sessions and prevent fatigue or overtraining. AI-assisted refereeing systems could enhance rule enforcement by detecting infractions such as line violations or improper serves, complementing existing technologies like Hawk-Eye.

It is also possible to integrate AI with augmented reality (AR), visualization, and immersive applications, such as VIRD (Lin et al., 2024), TIVEE (Chu et al., 2022), and TacticFlow (Wu et al., 2022), with the aim of enhancing immersive match analysis and realistic training simulations. In this regard, we highlight the Coach AI Badminton Project from the Advanced Database System Laboratory at National Yang Ming Chiao Tung University (CoachAI Research Team 2024), which applies AI for tactical analysis, while also exploring visualization techniques to represent and display both microscopic and macroscopic competition data for easier understanding, stroke and movement forecasting, as well as the ShuttleSet dataset. Additionally, AI could contribute to analyzing psychological and emotional data by interpreting facial expressions or vocal variations, providing insights into players' mental states and suggesting strategies to manage stress during matches. Fan engagement, though less explored in badminton, also presents significant opportunities using AI-driven tools like personalized content recommendations, virtual reality experiences, and real-time game analytics, fostering deeper connections between audiences and the sport. Together, these applications illustrate the transformative potential of AI in badminton, enhancing every aspect of the game from performance to spectator experience.

As AI-based tools continue to evolve, their adoption in badminton research also requires attention to practical and ethical considerations. Models trained primarily on elite-level data may not generalize well to amateur or developmental contexts, underscoring the importance of validating systems under the specific conditions in which they will be used. Likewise, the growing use of video and wearable sensors reinforces the need for clear procedures regarding consent, data protection and responsible handling of athlete information. Addressing these issues helps ensure that technological advances remain aligned with fair, transparent and context-appropriate applications.

At the same time, broader structural challenges also need to be acknowledged. Many of the approaches discussed in this paper rely on high-quality datasets, specialized equipment and technical expertise that are not equally accessible across all training environments. This highlights the importance of data democratization and of developing tools that coaches and practitioners with limited resources can adopt (whether through simpler interfaces, lower-cost acquisition methods, or open datasets that support evidence-based practice beyond elite settings). Expanding access in this way can help ensure that the benefits of AI are not restricted to highly resourced teams, but become practical and usable across the wider badminton community.

Finally, it is worth mentioning that badminton data analysis is inherently multidisciplinary, and must involve effective collaboration between professionals from different areas to integrate diverse perspectives and expertise. Statisticians, data scientists, coaches, physiologists, and professionals in sports science must work together to develop effective approaches. In this paper, we aim to leverage the collaboration between these professionals, by discussing problems, techniques, challenges, limitations, and potential themes for future research. By providing structured, actionable insights, the techniques discussed in this paper empower coaches and players to make well-informed, strategic decisions, ultimately raising the quality of performance in badminton and providing a scientific edge in competitive AI. This supportive role aligns with the overarching goal of sports analytics: to assist, not replace, human expertise.

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